

Mathematics A

Unit 2 • Chapter OL-T38 Classified

Cross-session • chapter-organised candidate

16 classified items • 80 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

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Contents

Unit 2 • Unit 2 • Chapter OL-T38 • 16 items • 80 marks

Chapter	Items	Marks	Starts
01. OL-T38 Differentiation	16	80	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Differentiation

Unit 2 • 80 marks • 16 item(s)

June 2025 | 4MA1-2H Q17 | 6 marks

Coordinates of polynomial stationary points

June 2024 | 4MA1-1H Q18 | 5 marks

Points where gradient equals a value

November 2023 | 4MA1-1H Q16 | 4 marks

Points where gradient equals a value

November 2023 | 4MA1-2H Q18 | 4 marks

Velocity and acceleration from displacement

June 2023 | 4MA1-1H Q11 | 6 marks

Points where gradient equals a value

November 2024 | 4MA1-1H Q20 | 4 marks

Equation of a calculus tangent

June 2022 | 4MA1-1H Q17 | 5 marks

Instantaneous rest or direction reversal

January 2022 | 4MA1-1H Q24 | 6 marks

Recover curve parameters from point and gradient data

January 2022 | 4MA1-2H Q24 | 5 marks

Instantaneous rest or direction reversal

January 2020 | 4MA1-1H Q15 | 5 marks

Maximum volume from geometric dimensions

January 2021 | 4MA1-1H Q17 | 5 marks

Maximum volume from geometric dimensions

January 2021 | 4MA1-2H Q19 | 5 marks

Instantaneous rest or direction reversal

June 2021 | 4MA1-2H Q21 | 5 marks

Recover curve parameters from point and gradient data

May-Nov 2020 | 1H Q23 | 4 marks

Points where gradient equals a value

January 2023 | 1H Q20 | 5 marks

Maximum volume from geometric dimensions

... 1 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

.....
(4)

(Total for Question 17 is 6 marks)

WORKING SPACE

Question 08

4MA1-1H Q18

MODULAR UNIT 2

5 marks

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

WORKING SPACE

Question 13

4MA1-1H Q16 demand

MODULAR UNIT 2

4 marks

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....

WORKING SPACE

Question 30

4MA1-2H Q18 whole

MODULAR UNIT 2

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

WORKING SPACE

Question 11

4MA1-1H Q11

MODULAR UNIT 2

6 marks

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

$$\dots\dots\dots$$

(4)

WORKING SPACE

Question 13

4MA1-1H Q20 M01

MODULAR UNIT 2

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

WORKING SPACE

Question 15

4MA1-1H Q17 Q17

MODULAR UNIT 2

5 marks

- 17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

WORKING SPACE

Question 15

4MA1-1H Q24

MODULAR UNIT 2

6 marks

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

 $y = \dots\dots\dots$

WORKING SPACE

Question 35

4MA1-2H Q24

MODULAR UNIT 2

5 marks

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

WORKING SPACE

15

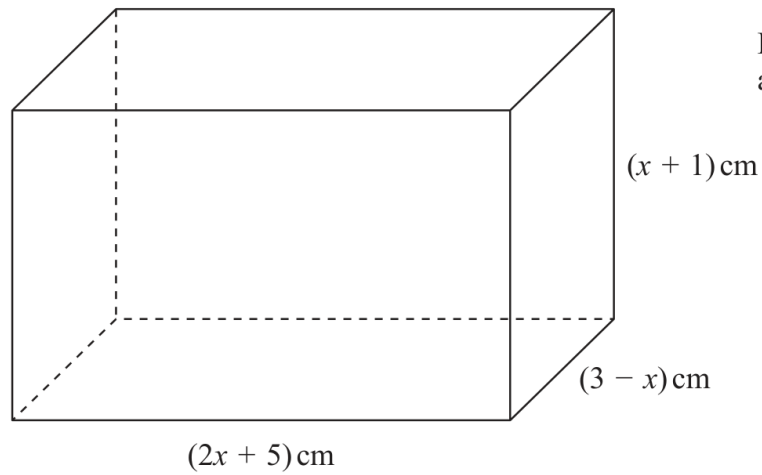


Diagram **NOT**
accurately drawn

The diagram shows a cuboid of volume $V \text{ cm}^3$

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

There is a value of x for which the volume of the cuboid is a maximum.

(b) Find this value of x .

Show your working clearly.

Give your answer correct to 3 significant figures.

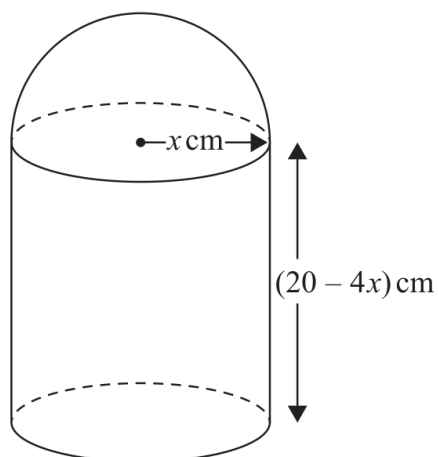
$x = \dots\dots\dots$

(5)

YOUR WORKING

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.



The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is $V \text{ cm}^3$ where $V = \frac{1}{3} \pi y$

Find the maximum value of y .
Show clear algebraic working.

YOUR WORKING

- 19 A particle P is moving along a straight line.
The fixed point O lies on this line.

At time t seconds where $t \geq 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

..... metres

YOUR WORKING

- 21** The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

$a = \dots\dots\dots$

YOUR WORKING

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

YOUR WORKING

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is V cm³

Find the maximum value of V

Give your answer correct to the nearest whole number.



YOUR WORKING

(Total for Question 20 is 5 marks)

YOUR WORKING

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

YOUR WORKING
